

Donghwan Rho

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Research Interests

- Implementing Language Models Under Homomorphic Encryption
- Mathematical Reasoning in Large Language Models
- Interpretability of Language Models

Education

Ph.D.	Seoul National University , Mathematical Sciences	Mar. 2021 – Present
	• Advisor: Ernest K. Ryu • Expected Graduation: Feb. 2026	
BS	Korea University , Mathematics Education	Mar. 2015 – Feb. 2021
	• Graduated 1st in Class • (Mandatory) Military Service: May 2017 - Jan. 2019 • Teacher's Certificate for Secondary Regular Teacher of Mathematics	

Experience

Workshop Coordinator , SNU-KAIST ML Theory Workshop	Gangneung, South Korea
• Assisted in planning the schedule, selecting the venue, and managing workshop programs.	Aug. 2024
Teaching Practicum	Seoul, South Korea
• Danggok high school	Apr. 2020 - May 2020
Military Service , Auxiliary Police	Seoul, South Korea
• Assisted in maintaining public order and security.	May 2017 – Jan. 2019

Publications

*: Equal Contribution

[3]	Studying the Korean Word-Chain Game with RLVR: Mitigating Reward Conflicts via Curriculum Learning Donghwan Rho In progress Paper	Oct. 2025
[2]	Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models Donghwan Rho* , Sieun Seo*, Hyewon Sung, Chohong Min, Ernest K. Ryu Submitted Paper	Oct. 2025
[1]	Encryption-Friendly LLM Architecture Donghwan Rho* , Taeseong Kim*, Minje Park, Jung Woo Kim, Hyunsik Chae, Ernest K. Ryu, Jung Hee Cheon International Conference on Learning Representations (ICLR) 2025 Paper Code	Oct. 2024

Talks & Presentations

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|---|-----------|
| Seoul National University Research Institute of Mathematics Symposium , Talk | Nov. 2025 |
| <ul style="list-style-type: none">Title: Implementing Efficient Language Models under Homomorphic Encryption
Link | |
| Korean Mathematical Society Annual Meeting , Talk | Oct. 2025 |
| <ul style="list-style-type: none">Title: Homomorphically encrypted language models stabilized with TSP-sorted tokens
Link | |
| SNU-Samsung Electronics Collaboration Symposium , Poster Presentation | Aug. 2024 |
| <ul style="list-style-type: none">Title: Encryption-Friendly LLM Architecture | |
| SNU-KAIST ML Theory Workshop , Poster Presentation | Aug. 2024 |
| <ul style="list-style-type: none">Title: Encryption-Friendly LLM Architecture | |

Honors and Awards

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| Outstanding TA , Seoul National University | July 2025 |
| Introduction to Computer Programming and Artificial Intelligence for Scientists | |
| University Students Contest of Mathematics , Silver Prize | Dec. 2018 |
| Academic Excellence Scholarship , Korea University | Aug. 2015 |

Teaching

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| Peer Tutoring in Basic Sciences , Lead Teaching Assistant | Mar. 2024 - Present |
| <ul style="list-style-type: none">Represented section TAs to ensure the smooth operation of the peer tutoring program.Oversaw tutorings in Mathematics, Physics, Chemistry, Biology, and Statistics. | |
| Basic Calculus , Teaching Assistant | Sep. 2021 - Present |
| <ul style="list-style-type: none">Matched tutors and tutees for the <i>Basic Calculus</i> peer tutoring program.Managed program logistics to ensure efficient operation. | |
| SNU Pre-College Program , Head TA, Instructor | |
| <ul style="list-style-type: none">Managed logistics of the mathematics program for freshman undergraduate.Gave lectures.Winter 2024, Winter 2023, Winter 2022 | |
| Introduction to Computer Programming and Artificial Intelligence for Scientists , Instructor | Spring 2025 |
| <ul style="list-style-type: none">English CourseSelected as an outstanding TA | |
| Topics in Machine Intelligence | Fall 2024 |
| <ul style="list-style-type: none">Topics: Continuous-depth neural networks, Diffusion Models, Large Language Models, Vision Language Models, State-space ModelsEnglish CourseGraduate Course | |
| Mathematical algorithms II | Fall 2022 |
| <ul style="list-style-type: none">Topics: Reinforcement Learning, Diffusion Models, Natural Language Processing | |

- English Course
- Graduate Course

Calculus 2 Practice, Instructor

- Fall 2021, Fall 2025

Calculus 1 Practice, Instructor

- Spring 2024, Fall 2023, Spring 2023, Spring 2022, Spring 2021

Technical Skills

Languages: Korean (Native), English (Fluent)

Programming Languages: Python, C++

Tools & Technologies: LaTeX, GitHub